

- Description

This module manages a liquid level monitoring system, ensuring that:

1. **The liquid level in a storage tank is continuously measured and checked against safety limits.**
2. **If the liquid level exceeds the maximum threshold, an overflow warning is triggered.**
3. **If the level is too low, a refill recommendation is generated to maintain operational stability.**
4. **A log of all level readings, warnings, and refill actions is maintained for safety compliance.**

This system helps prevent liquid overflow incidents and ensures a stable supply in industrial applications.

- Specification

module LiquidLevelMonitoring

var currentLevel, maxLevel, minLevel: real

var levelLog: sequence of string

process updateLevel(changeAmount: real)

functional scenario form:

currentLevel <= maxLevel && levelLog = levelLog $\oplus$ ("Level Updated: " + currentLevel + " L"), Liquid level updated successfully.

||

currentLevel > maxLevel && levelLog = levelLog $\oplus$ ("Overflow Warning: " + currentLevel + " L"), Overflow warning triggered.

||

currentLevel < minLevel && levelLog = levelLog $\oplus$ ("Low Level Alert: " + currentLevel + " L"), Low level alert triggered.

end process

end module

- Java Code

```
package code;
import java.util.*;
class LiquidTank {
    private double maxLevel;
    private double minLevel;
    private double currentLevel;
    private List<String> levelLog;
    public LiquidTank(double maxLevel, double minLevel, double initialLevel) {
        this.maxLevel = maxLevel;
        this.minLevel = minLevel;
        this.currentLevel = initialLevel;
    }
}
```

```

        this.levelLog = new ArrayList<>();
    }
    public void updateLevel(double changeAmount) {
        currentLevel += changeAmount;
        String message = String.format("Current Level: %.1f liters", currentLevel);
        if (currentLevel > maxLevel) {
            message += " | WARNING: Overflow Risk!";
        } else if (currentLevel < minLevel) {
            message += " | ALERT: Low Level! Refill Recommended.";
        }
        System.out.println(message);
        levelLog.add(message);
    }
    public void printLevelLog() {
        System.out.println("\nLiquid Level Log:");
        for (String log : levelLog) {
            System.out.println(log);
        }
    }
}

public class c27_LiquidLevelSystem {
    public static void main(String[] args) {
        LiquidTank tank = new LiquidTank(1000.0, 200.0, 500.0);
        // testcase-VT:
        tank.updateLevel(-47.70);
        tank.updateLevel(419.47);
        tank.updateLevel(340.35);
        tank.updateLevel(-394.61);
        tank.updateLevel(-84.99);
        tank.updateLevel(-305.24);
        tank.updateLevel(156.97);
        tank.updateLevel(-472.42);
        tank.updateLevel(-438.21);
        tank.updateLevel(437.73);
        tank.updateLevel(-354.21);
        tank.updateLevel(72.38);
        tank.updateLevel(147.96);
        tank.updateLevel(104.37);
        tank.updateLevel(-20.31);
        tank.updateLevel(-388.59);
        tank.updateLevel(-114.67);
        tank.updateLevel(174.98);
        tank.updateLevel(-160.65);
    }
}

```

```
tank.updateLevel(289.97);
tank.updateLevel(269.40);
tank.updateLevel(178.54);
tank.updateLevel(-104.10);
tank.updateLevel(200.84);
tank.updateLevel(-229.45);
tank.updateLevel(-17.38);
tank.updateLevel(-70.57);
tank.updateLevel(-345.08);
tank.updateLevel(425.23);
tank.updateLevel(-414.21);
tank.updateLevel(-498.72);
tank.updateLevel(-425.69);
tank.updateLevel(-438.79);
tank.updateLevel(455.23);
tank.updateLevel(-255.52);
tank.updateLevel(-227.24);
tank.updateLevel(-252.20);
tank.updateLevel(264.10);
tank.updateLevel(33.22);
tank.updateLevel(-223.43);
tank.updateLevel(-25.20);
tank.updateLevel(-441.32);
tank.updateLevel(220.57);
tank.updateLevel(376.25);
tank.updateLevel(281.18);
tank.updateLevel(122.68);
tank.updateLevel(280.63);
tank.updateLevel(218.11);
tank.updateLevel(312.72);
tank.updateLevel(398.77);
tank.updateLevel(-181.28);
tank.updateLevel(230.93);
tank.updateLevel(-333.42);
tank.updateLevel(-20.91);
tank.updateLevel(-488.63);
tank.updateLevel(-463.68);
tank.updateLevel(-233.82);
tank.updateLevel(-484.67);
tank.updateLevel(-480.63);
tank.updateLevel(426.35);
tank.updateLevel(170.62);
tank.updateLevel(256.56);
tank.updateLevel(43.13);
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
00000000000000000000000000000000000000000000000000000000000000);  
//      tank.updateLevel(-  
17976931348606807000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000);  
//      tank.updateLevel(-  
17976931348623157000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000);  
//      tank.updateLevel(-  
17976931348623157000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000)  
//      tank.updateLevel(-  
17976931348623157000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000);  
  
    // testcase-Z3:  
//      tank.updateLevel(-47.70);  
//      tank.updateLevel(419.47);  
//      tank.updateLevel(340.35);  
//      tank.updateLevel(-394.61);  
//      tank.updateLevel(-84.99);  
//      tank.updateLevel(-305.24);  
//      tank.updateLevel(156.97);  
//      tank.updateLevel(-472.42);  
//      tank.updateLevel(-438.21);  
//      tank.updateLevel(437.73);  
//      tank.updateLevel(-354.21);  
//      tank.updateLevel(72.38);  
//      tank.updateLevel(147.96);  
//      tank.updateLevel(104.37);  
//      tank.updateLevel(-20.31);  
//      tank.updateLevel(-388.59);
```

```
//      tank.updateLevel(-114.67);
//      tank.updateLevel(174.98);
//      tank.updateLevel(-160.65);
//      tank.updateLevel(289.97);
//      tank.updateLevel(269.40);
//      tank.updateLevel(178.54);
//      tank.updateLevel(-104.10);
//      tank.updateLevel(200.84);
//      tank.updateLevel(-229.45);
//      tank.updateLevel(-17.38);
//      tank.updateLevel(-70.57);
//      tank.updateLevel(-345.08);
//      tank.updateLevel(425.23);
//      tank.updateLevel(-414.21);
//      tank.updateLevel(-498.72);
//      tank.updateLevel(-425.69);
//      tank.updateLevel(-438.79);
//      tank.updateLevel(455.23);
//      tank.updateLevel(-255.52);
//      tank.updateLevel(-227.24);
//      tank.updateLevel(-252.20);
//      tank.updateLevel(264.10);
//      tank.updateLevel(33.22);
//      tank.updateLevel(-223.43);
//      tank.updateLevel(-25.20);
//      tank.updateLevel(-441.32);
//      tank.updateLevel(220.57);
//      tank.updateLevel(376.25);
//      tank.updateLevel(281.18);
//      tank.updateLevel(122.68);
//      tank.updateLevel(280.63);
//      tank.updateLevel(218.11);
//      tank.updateLevel(312.72);
//      tank.updateLevel(398.77);
//      tank.updateLevel(-181.28);
//      tank.updateLevel(230.93);
//      tank.updateLevel(-333.42);
//      tank.updateLevel(-20.91);
//      tank.updateLevel(-488.63);
//      tank.updateLevel(-463.68);
//      tank.updateLevel(-233.82);
//      tank.updateLevel(-484.67);
//      tank.updateLevel(-480.63);
//      tank.updateLevel(426.35);
```

```
//      tank.updateLevel(170.62);
//      tank.updateLevel(256.56);
//      tank.updateLevel(43.13);
//      tank.updateLevel(-343.52);
//      tank.updateLevel(118.66);
//      tank.updateLevel(474.40);
//      tank.updateLevel(-204.33);
//      tank.updateLevel(314.81);
//      tank.updateLevel(64.80);
//      tank.updateLevel(67.03);
//      tank.updateLevel(-181.06);
//      tank.updateLevel(390.08);
//      tank.updateLevel(18.97);
//      tank.updateLevel(345.67);
//      tank.updateLevel(-433.96);
//      tank.updateLevel(-135.57);
//      tank.updateLevel(-321.85);
//      tank.updateLevel(-180.88);
//      tank.updateLevel(-129.64);
//      tank.updateLevel(-376.48);
//      tank.updateLevel(-465.26);
//      tank.updateLevel(-4.81);
//      tank.updateLevel(492.35);
//      tank.updateLevel(421.69);
//      tank.updateLevel(-435.78);
//      tank.updateLevel(411.95);
//      tank.updateLevel(-473.94);
//      tank.updateLevel(203.66);
//      tank.updateLevel(-185.68);
//      tank.updateLevel(-155.07);
//      tank.updateLevel(307.43);
//      tank.updateLevel(-328.31);
//      tank.updateLevel(90.71);
//      tank.updateLevel(331.06);
//      tank.updateLevel(95.49);
//      tank.updateLevel(63.98);
//      tank.updateLevel(-103.68);
//      tank.updateLevel(104.26);
//      tank.updateLevel(-160.39);
//      tank.updateLevel(-261.78);

// testcase-UVT:
tank.updateLevel(-47.70);
tank.updateLevel(419.47);
```

```
tank.updateLevel(340.35);
tank.updateLevel(-394.61);
tank.updateLevel(-84.99);
tank.updateLevel(-305.24);
tank.updateLevel(156.97);
tank.updateLevel(-472.42);
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tank.updateLevel(-114.67);
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tank.updateLevel(-160.65);
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tank.updateLevel(264.10);
tank.updateLevel(33.22);
tank.updateLevel(-223.43);
tank.updateLevel(-25.20);
tank.updateLevel(-441.32);
tank.updateLevel(220.57);
tank.updateLevel(376.25);
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```

```
tank.updateLevel(280.63);
tank.updateLevel(218.11);
tank.updateLevel(312.72);
tank.updateLevel(398.77);
tank.updateLevel(-181.28);
tank.updateLevel(230.93);
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tank.updateLevel(-233.82);
tank.updateLevel(-484.67);
tank.updateLevel(-480.63);
tank.updateLevel(426.35);
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tank.updateLevel(-155.07);
```

```

        tank.updateLevel(307.43);
        tank.updateLevel(-328.31);
        tank.updateLevel(90.71);
        tank.updateLevel(331.06);
        tank.updateLevel(95.49);
        tank.updateLevel(63.98);
        tank.updateLevel(-103.68);
        tank.updateLevel(104.26);
        tank.updateLevel(-160.39);
        tank.updateLevel(-261.78);

//      tank.printLevelLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c27_LiquidLevelSystem.java	66.7 %	4	2	6
> LiquidTank	66.7 %	4	2	6
> c27_LiquidLevelSystem		0	0	0

3. Vibration testing instruction coverage: 94.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.1 %	370	17,553	17,923
▼ src	2.1 %	370	17,553	17,923
▼ code	2.1 %	370	17,553	17,923
▼ c27_LiquidLevelSystem.java	94.4 %	370	22	392
> LiquidTank	76.5 %	62	19	81
> c27_LiquidLevelSystem	99.0 %	308	3	311

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c27_LiquidLevelSystem.java	83.3 %	5	1	6
> LiquidTank	83.3 %	5	1	6
> c27_LiquidLevelSystem		0	0	0

5. Fuzz testing instruction coverage: 99.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.2 %	391	17,529	17,920
▼ src	2.2 %	391	17,529	17,920
▼ code	2.2 %	391	17,529	17,920
▼ c27_LiquidLevelSystem.java	99.2 %	391	3	394
> c27_LiquidLevelSystem	99.0 %	310	3	313
> LiquidTank	100.0 %	81	0	81

6. Z3 branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c27_LiquidLevelSystem.java	66.7 %	4	2	6
> LiquidTank	66.7 %	4	2	6
> c27_LiquidLevelSystem		0	0	0

7. Z3 instruction coverage: 96.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	3.0 %	670	21,728	22,398
▼ src	3.0 %	670	21,728	22,398
▼ code	3.0 %	670	21,728	22,398
▼ c27_LiquidLevelSystem.java	96.8 %	670	22	692
> LiquidTank	76.5 %	62	19	81
> c27_LiquidLevelSystem	99.5 %	608	3	611

8. Unimproved V-method branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c27_LiquidLevelSystem.java	66.7 %	4	2	6
> LiquidTank	66.7 %	4	2	6
> c27_LiquidLevelSystem		0	0	0

9. Unimproved V-method instruction coverage: 96.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	3.0 %	670	21,728	22,398
▼ src	3.0 %	670	21,728	22,398
▼ code	3.0 %	670	21,728	22,398
▼ c27_LiquidLevelSystem.java	96.8 %	670	22	692
> LiquidTank	76.5 %	62	19	81
> c27_LiquidLevelSystem	99.5 %	608	3	611